

power - practice problems

name _____

1. A power bill charges for kilowatt hours. Is a kilowatt hour a unit of power or a unit of energy?
2. A small oil-filled radiator heater is warming a room. This type of heater takes longer to heat up, but it provides quiet, steady warmth after it gets hot. If it transfers 1800 J of energy to the air in 3.0 s, what is its power use?
3. A phone charger with a 1.8 m cable transfers energy at 12 W for 15 minutes while a phone is below 20% battery. How many joules of energy does it transfer?
4. A 1500 W space heater runs under a desk for 2.5 hours on a cold morning. How much energy does it use in kWh?

5. Electricity costs \$0.18 for each kWh. How much does the 3.75 kWh from the heater in the previous example cost?

6. A roof solar panel is about 1.7 m tall and 1.0 m wide. On a clear day it averages 420 W for 6.0 hours. How much energy does it produce in kWh?

7. A 2000 kg electric car accelerates from rest to 27 m/s. Its motor delivers 150 kW of power. Ignoring losses, how long does it take?

8. A device runs for 12 minutes on a 9.0 V battery. How much energy does it use?

9. A 12 V LED strip under a shelf draws 0.75 A. The strip is 2.0 m long. How much power does it use, and how much energy does it convert in 5.0 minutes?

10. A 60 W light bulb is connected to 120 V. What current does it draw?

11. A laptop charger label says it can accept 100-240 V from the wall. On its output side, it provides 65 W at 20 V. What current does it provide to the laptop?

12. The resistance of a resistor stays the same. If the voltage across it doubles, what happens to the power dissipated by the resistor?

13. A $30\ \Omega$ resistor has 0.20 A through it. How much power does it dissipate?

14. A $240\ \Omega$ resistor has 12 V across it during a lab test. The resistor is rated for 0.25 W, but first calculate the power it would dissipate.

15. A 10 W LED bulb gives the same light as a 60 W incandescent bulb. A lamp is on for 5.0 hours a day for a year. At \$0.15 per kWh, how much money does the LED bulb save?

16. A small cup-warmer plate should use 48 W when it's connected to 24 V. What resistance should its heating element have?

17. A 12 V motor rated for 36 W spins the three-blade fan in a desk cooler. What current does it draw, and what is its effective resistance while it's running?

18. A 150 Ω resistor is connected across 9.0 V for 10 minutes in a simple timing circuit. How much energy does it convert to heat?